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Hybrid filter, multiplexer, radio-frequency module, and communication device

Abstract

A hybrid filter includes a module substrate having a major surface and a major surface that are opposite to each other, acoustic wave resonator elements disposed at the module substrate, inductors disposed at the module substrate, and a capacitor disposed at the module substrate. The pass band of the hybrid filter is wider than the resonance band width of the acoustic wave resonator elements. One of the acoustic wave resonator elements, the inductors, and the capacitor is a first circuit element, and the first circuit element is disposed at the major surface. Another of the acoustic wave resonator elements, the inductors, and the capacitor is a second circuit element, and the second circuit element is disposed at the major surface.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a bypass continuation of PCT filing PCT/JP2021/029237, filed on Aug. 5, 2021, which claims priority to Japanese Patent Application 2020-206045, filed on Dec. 11, 2020, and Japanese Patent Application 2021-059093, filed on Mar. 31, 2021, the entire contents of all of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a hybrid filter, a multiplexer, a radio-frequency module, and a communication device.

BACKGROUND ART

(2) Patent Document 1 discloses a hybrid acoustic LC filter including an acoustic resonator (an acoustic wave resonator), an inductor, and a capacitor. According to Patent Document 1, this hybrid acoustic LC filter provides a relatively wide pass band and at the same time achieves a strict out-of-band rejection specification.

CITATION LIST

Patent Document

(3) Patent Document 1: Japanese Unexamined Patent Application Publication No. 2020-14204

SUMMARY OF DISCLOSURE

Technical Problem

(4) However, because the hybrid acoustic LC filter disclosed in Patent Document 1 is formed by an acoustic wave resonator-inductor-capacitor combination, the hybrid acoustic LC filter includes many components. As a result, a problem arises in which the hybrid acoustic LC filter is relatively large in size. If the mounting density is increased for the purpose of reducing the size, another problem arises in which unnecessary coupling between components occurs, and the bandpass characteristic of the filter deteriorates.

(5) The present disclosure has been made to address the problems described above, and an object thereof is to provide a small hybrid filter including an acoustic wave resonator, an inductor, and a capacitor in which degradation of filter characteristics is suppressed and further provide a multiplexer, a radio-frequency module, and a communication device.

Solution to Problem

(6) A hybrid filter according to an aspect of the present disclosure includes a substrate having a first major surface and a second major surface that are opposite to each other, one or more acoustic wave resonator elements disposed at the substrate, one or more first inductors disposed at the substrate, and one or more first capacitors disposed at the substrate. The pass band width of the hybrid filter is wider than the resonance band width of the one or more acoustic wave resonator elements. One of the one or more acoustic wave resonator elements, the one or more first inductors, and the one or more first capacitors is a first circuit element, and the first circuit element is disposed at the first major surface. Another of the one or more acoustic wave resonator elements, the one or more first inductors, and the one or more first capacitors is a second circuit element, and the second circuit element is disposed at the second major surface.

Advantageous Effects

(7) The present disclosure provides a small hybrid filter including an acoustic wave resonator element, an inductor, and a capacitor in which degradation of filter characteristics is suppressed and further provide a multiplexer, a radio-frequency module, and a communication device.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a circuit configuration diagram of a radio-frequency module and a communication

device according to an embodiment.

(2) FIG. 2A illustrates an example of a circuit configuration of a first hybrid filter according to the embodiment.

(3) FIG. 2B illustrates an example of a circuit configuration of a second hybrid filter according to the embodiment.

(4) FIG. 2C illustrates an example of a circuit configuration of an LC filter according to the embodiment.

(5) FIG. 3A provides schematic plan views of a configuration of a radio-frequency module according to a first example.

(6) FIG. 3B is a schematic sectional view of the configuration of the radio-frequency module according to the first example.

(7) FIG. 4 provides schematic plan views of a configuration of a radio-frequency module according to a second example.

(8) FIG. 5 provides schematic plan views of a configuration of a radio-frequency module according to a third example.

(9) FIG. 6 is a schematic sectional view of a configuration of a radio-frequency module according to a fourth example.

(10) FIG. 7A is a schematic partial plan view of a configuration of a radio-frequency module according to a first modification.

(11) FIG. 7B is a schematic partial plan view of a configuration of a radio-frequency module according to a second modification.

(12) FIG. 7C is a schematic partial plan view of a configuration of a radio-frequency module according to a third modification.

(13) FIG. 8 is a circuit configuration diagram of a radio-frequency module and a communication device according to a fifth example.

DESCRIPTION OF EMBODIMENTS

(14) Hereinafter, an embodiment of the present disclosure will be described in detail. It should be noted that the embodiment described below is a comprehensive or specific instance. Specifics including numerical values, shapes, materials, constituent elements, arrangements of the constituent elements, and modes of connection given in the following embodiment are mere instances and are not intended to limit the present disclosure. Among the constituent elements in the following examples and modifications, constituent elements not recited in any of the independent claims are described as arbitrary constituent elements. The sizes or size ratios of the constituent elements illustrated in the drawings are not necessarily exact. Like reference symbols are used to denote substantially like configurations in the drawings, and redundant descriptions thereof may be omitted or simplified.

(15) In the following description, words used to express relationships between elements, such as parallel and vertical, words used to express the shape of an element, such as rectangular, and numerical ranges do not necessarily denote the exact meanings but denote substantially the same meanings involving, for example, several percent differences.

(16) As used herein, the expression “A is disposed at a first major surface of a substrate” means not only that A is mounted directly on the first major surface but also that of the space on the first major surface side and the space on the second major surface side that are separated by the substrate, A is disposed in the space on the first major surface side. Specifically, the expression also means that A is mounted at the first major surface with another circuit element, an electrode, or another element that is interposed therebetween.

(17) In the following description, the term “couple” includes not only the case in which a circuit component is coupled directly to another circuit component by using a connection terminal and/or an interconnection conductor but also the case in which a circuit component is electrically coupled to another circuit component via still another circuit component. The expression “coupled between

A and B” means that a circuit element is coupling to both A and B while the circuit element is positioned between A and B.

(18) In the drawings described below, the x axis and the y axis are axes perpendicular to each other in a plane parallel to major surfaces of a module substrate. The z axis is an axis perpendicular to major surfaces of a module substrate; the forward direction of the z axis indicates an upward direction, and the reverse direction of the z axis indicates a downward direction.

(19) Regarding the module configuration of the present disclosure, the expression “plan view” denotes that an object orthogonally projected on an xy plane is viewed from the front side in the positive direction of the z axis. The expression “a component is disposed at a major surface of a substrate” includes the case in which the component is positioned in contact with the major surface of the substrate, the case in which the component is positioned over the major surface without contact with the major surface, and the case in which the component is partially embedded in the substrate at the major surface.

(20) In the following description, in the case in which A, B and C is mounted on the substrate, the expression “when the substrate (or the major surface of the substrate) is viewed in plan view, C is disposed between A and B” means that the region occupied by C is intersected by at least one of the line segments connecting points within A and points within B when the substrate is viewed in plan view. The plan view of a substrate denotes that the substrate and circuit elements mounted on the substrate are viewed in the state in which the substrate and circuit elements are orthogonally projected on a plane parallel to the major surface of the substrate.

(21) In the following, a “transmit path” refers to a transfer line formed by, for example, an interconnection for transferring radio-frequency transmit signals, an electrode directly coupled to the interconnection, and a terminal directly coupled to the interconnection or electrode. Similarly, a “receive path” refers to a transfer line formed by, for example, an interconnection for transferring radio-frequency receive signals, an electrode directly coupled to the interconnection, and a terminal directly coupled to the interconnection or electrode.

Embodiment

(22) [1. Configuration of Radio-Frequency Module **1** and Communication Device **5** According to Embodiment]

(23) FIG. **1** is a circuit configuration diagram of a radio-frequency module **1** and a communication device **5** according to the embodiment. As illustrated in this drawing, the communication device **5** includes the radio-frequency module **1**, an antenna **2**, a radio-frequency (RF) signal processing circuit (RFIC) **3**, and a baseband signal processing circuit (BBIC) **4**.

(24) The RFIC **3** is a radio frequency signal processing circuit configured to process a radio-frequency signal received or to be transmitted by the antenna **2**.

(25) Specifically, the RFIC **3** processes a receive signal inputted through a receive path of the radio-frequency module **1** by for example, downconversion and outputs the receive signal generated by the signal processing to the BBIC **4**. The RFIC **3** outputs a radio-frequency transmit signal processed based on a signal inputted from the BBIC **4** to a transmit path of the radio-frequency module **1**.

(26) The BBIC **4** performs data processing using signals of frequencies lower than radio-frequency signals transferred in the radio-frequency module **1**. The signal processed by the BBIC **4** is used as, for example, an image signal for displaying an image or a sound signal for talk through a speaker.

(27) The RFIC **3** functions as a controller for controlling connections of switches **50** and **51** included in the radio-frequency module **1** based on whether the radio-frequency module **1** is used for transmission or reception and which communication band (frequency range) is used. Specifically, the RFIC **3** controls connections of the switches **50** and **51** included in the radio-frequency module **1** by using a control signal (not illustrated in the drawing). The controller may be external to the RFIC **3**; for example, the controller may be provided in the radio-frequency module **1** or the BBIC **4**.

(28) The RFIC **3** also functions as a controller for controlling the gain of power amplifiers **30** and **31** included in the radio-frequency module **1** and a supply voltage V_{cc} and a bias voltage V_{bias} that are to be supplied to the power amplifiers **30** and **31**.

(29) The antenna **2** is coupled to an antenna connection terminal **100** of the radio-frequency module **1**. The antenna **2** emits a radio-frequency signal outputted from the radio-frequency module **1**. The antenna **2** also receives a radio-frequency signal from outside and outputs the radio-frequency signal to the radio-frequency module **1**.

(30) In the communication device **5** according to the present embodiment, the antenna **2** and the BBIC **4** are non-essential constituent elements.

(31) The following describes a detailed configuration of the radio-frequency module **1**.

(32) As illustrated in FIG. **1**, the radio-frequency module **1** includes the antenna connection terminal **100**, receive output terminals **120** and **121**, transmit input terminals **130** and **131**, hybrid filters **10** and **11**, the switches **50** and **51**, filters **15**, **16**, **17**, and **18**, matching circuits **45**, **46**, **47**, and **48**, low-noise amplifiers **20** and **21**, and the power amplifiers **30** and **31**.

(33) The antenna connection terminal **100** is an antenna common terminal coupled to the antenna **2**.

(34) The hybrid filters **10** and **11** are respectively an example of a first hybrid filter and an example of a second hybrid filter. The hybrid filters **10** and **11** each include one or more acoustic wave resonator elements, one or more inductors, and one or more capacitors. One terminal of the hybrid filter **10** is coupled to the antenna connection terminal **100**, and the other terminal is coupled to the switch **50**. One terminal of the hybrid filter **11** is coupled to the antenna connection terminal **100**, and the other terminal is coupled to the switch **51**. The hybrid filters **10** and **11** form a multiplexer **90**.

(35) The acoustic wave resonator element is, for example, an acoustic wave resonator element using surface acoustic waves (SAWs) or an acoustic wave resonator element using bulk acoustic waves (BAWs).

(36) It is sufficient that the multiplexer **90** is configured to separate at least two different frequency ranges (communication bands) of radio-frequency signals. According to this, the number of filters forming the multiplexer **90** may be two or more.

(37) FIG. **2A** illustrates an example of a circuit configuration of the hybrid filter **10** according to the embodiment. FIG. **2B** illustrates an example of a circuit configuration of the hybrid filter **11** according to the embodiment.

(38) As illustrated in FIG. **2A**, the hybrid filter **10** includes acoustic wave resonator elements **P1** and **P2**, a capacitor **C3**, and inductors **L1**, **L2**, and **L3**.

(39) The inductor **L3** and the capacitor **C3** form an LC parallel resonant circuit. A series circuit composed of the acoustic wave resonator element **P1** and the inductor **L1** is provided between the ground and a node in a path connecting an input-output terminal **101** and the LC parallel resonant circuit. A series circuit composed of the acoustic wave resonator element **P2** and the inductor **L2** is provided between the ground and a node in a path connecting an input-output terminal **102** and the LC parallel resonant circuit. The acoustic wave resonator elements **P1** and **P2** form an acoustic wave resonator **A1**, and the acoustic wave resonator elements **P1** and **P2** are, for example, integrated into one chip. The expression “a plurality of acoustic wave resonator elements are integrated into one chip” is defined such that a plurality of acoustic wave resonator elements are formed on one piezoelectric substrate, or a plurality of acoustic wave resonator elements are included in one package.

(40) In the configuration described above, the pass band and attenuation band of the hybrid filter **10** are determined by controlling the resonant frequency of the LC parallel resonant circuit composed of the inductor **L3** and the capacitor **C3** and the resonant frequency and anti-resonant frequency of the acoustic wave resonator elements **P1** and **P2**. The LC parallel resonant circuit composed of the inductor **L3** and the capacitor **C3** determines the pass band of the hybrid filter **10**, and the acoustic wave resonator elements **P1** and **P2** determine attenuation poles.

(41) As illustrated in FIG. 2B, the hybrid filter **11** has a circuit configuration similar to the hybrid filter **10**, including acoustic wave resonator elements **P5** and **P6**, a capacitor **C4**, and inductors **L4**, **L5**, and **L6**.

(42) The inductor **L4** and the capacitor **C4** form an LC parallel resonant circuit. A series circuit composed of the acoustic wave resonator element **P5** and the inductor **L5** is provided between the ground and a node in a path connecting an input-output terminal **103** and the LC parallel resonant circuit. A series circuit composed of the acoustic wave resonator element **P6** and the inductor **L6** is provided between the ground and a node in a path connecting an input-output terminal **104** and the LC parallel resonant circuit. The acoustic wave resonator elements **P5** and **P6** form an acoustic wave resonator **A2**, and the acoustic wave resonator elements **P5** and **P6** are, for example, integrated into one chip.

(43) In the configuration described above, the pass band and attenuation band of the hybrid filter **11** are determined by controlling the resonant frequency of the LC parallel resonant circuit composed of the inductor **L4** and the capacitor **C4** and the resonant frequency and anti-resonant frequency of the acoustic wave resonator elements **P5** and **P6**. The LC parallel resonant circuit composed of the inductor **L4** and the capacitor **C4** determines the pass band of the hybrid filter **11**, and the acoustic wave resonator elements **P5** and **P6** determine attenuation poles.

(44) As a result, the hybrid filters **10** and **11**, by using LC circuits, achieve wide pass bands that cannot be achieved with acoustic wave resonator elements, and by using acoustic wave resonator elements, provide steep attenuation slopes that cannot be achieved with LC circuits.

(45) According to this, the pass band width of the hybrid filter **10** is wider than the resonance band width of the acoustic wave resonator elements **P1** and **P2**. Similarly, the pass band width of the hybrid filter **11** is wider than the resonance band width of the acoustic wave resonator elements **P5** and **P6**.

(46) In the present embodiment, the resonance band width of an acoustic wave resonator element is defined as the difference between the anti-resonant frequency and resonant frequency of the acoustic wave resonator element. A fractional resonance band width is defined as the proportion obtained by dividing the resonance band width by the mean value between the anti-resonant frequency and the resonant frequency. A general SAW or BAW resonator element is known to have a frequency range of 0.1 to 10 GHz and a 3 to 4% fractional resonance band width.

(47) The circuit configuration of the hybrid filter according to the present embodiment is not limited to the circuit configuration of the hybrid filters **10** and **11** described above. It is sufficient that the hybrid filter according to the present embodiment includes one or more acoustic wave resonator elements, one or more inductors, and one or more capacitors; and the pass band width of the hybrid filter is wider than the resonance band width of the acoustic wave resonator element. In the circuit configuration of the hybrid filter according to the present embodiment, no switch is provided between any acoustic wave resonator element and the LC circuit. For example, in the hybrid filter **10**, no switch is inserted between the LC parallel resonant circuit composed of the inductor **L3** and the capacitor **C3** and the acoustic wave resonator element **P1** and between the LC parallel resonant circuit and the acoustic wave resonator element **P2**.

(48) The filter **15** is a receive filter coupled between the switch **50** and the matching circuit **45**. The filter **16** is a transmit filter coupled between the switch **50** and the matching circuit **46**. The filter **17** is a receive filter coupled between the switch **51** and the matching circuit **47**. The filter **18** is a transmit filter coupled between the switch **51** and the matching circuit **48**. The filters **15** to **18** each may be an LC filter including an inductor and a capacitor.

(49) FIG. 2C illustrates an example of a circuit configuration of the filter **15** according to the embodiment. As illustrated in FIG. 2C, the filter **15** includes capacitors **C6**, **C7**, **C8**, and **C9** and inductors **L7** and **L8**. The filters **16** to **18** have the same circuit configuration as the filter **15**.

(50) The inductor **L7** and the capacitor **C7** form an LC parallel resonant circuit. The capacitor **C6** is provided in series in the path connecting an input-output terminal **105** and the LC parallel resonant

circuit. A series resonant circuit composed of the capacitor **C8** and the inductor **L8** is provided between the ground and a node in a path connecting an input-output terminal **105** and the LC parallel resonant circuit. The capacitor **C9** is provided between the ground and a node in a path connecting an input-output terminal **106** and the LC parallel resonant circuit.

(51) In the configuration described above, the pass band and attenuation band of the filter **15** are determined by controlling the resonant frequency of the LC parallel resonant circuit composed of the inductor **L7** and the capacitor **C7** and the resonant frequency of the LC series resonant circuit composed of the inductor **L8** and the capacitor **C8**.

(52) As a result, the filter **15**, by using an LC parallel resonant circuit and an LC series resonant circuit, achieves a wide pass band and also provides high attenuation in the attenuation band that is far from the pass band.

(53) The circuit configuration of the filters **15** to **18** according to the present embodiment is not limited to the circuit configuration of the filter **15** described above. It is sufficient that the filters **15** to **18** according to the present embodiment includes one or more inductors and one or more capacitors without having any acoustic wave resonator element.

(54) The low-noise amplifier **20** amplifies receive signals in a first communication band with low noise and outputs the receive signals to the receive output terminal **120**. The low-noise amplifier **21** amplifies receive signals in a second communication band with low noise and outputs the receive signals to the receive output terminal **121**.

(55) The power amplifier **30** amplifies transmit signals in the first communication band inputted from the transmit input terminal **130**. The power amplifier **31** amplifies transmit signals in the second communication band inputted from the transmit input terminal **131**.

(56) The matching circuit **45** is coupled between the low-noise amplifier **20** and the filter **15**. The matching circuit **45** provides impedance matching between the low-noise amplifier **20** and the filter **15**. The matching circuit **46** is coupled between the power amplifier **30** and the filter **16**. The matching circuit **46** provides impedance matching between the power amplifier **30** and the filter **16**. The matching circuit **47** is coupled between the low-noise amplifier **21** and the filter **17**. The matching circuit **47** provides impedance matching between the low-noise amplifier **21** and the filter **17**. The matching circuit **48** is coupled between the power amplifier **31** and the filter **18**. The matching circuit **48** provides impedance matching between the power amplifier **31** and the filter **18**.

(57) The switch **50** has a common terminal and two selection terminals. The common terminal of the switch **50** is coupled to the hybrid filter **10**. One of the selection terminals of the switch **50** is coupled to the filter **15**, and the other of the selection terminals of the switch **50** is coupled to the filter **16**. Specifically, the switch **50** is a time division duplex (TDD) switch for selectively connecting the hybrid filter **10** to the low-noise amplifier **20** or the power amplifier **30**. The switch **50** is implemented by, for example, a single-pole double-throw (SPDT) switching circuit.

(58) The switch **51** has a common terminal and two selection terminals. The common terminal of the switch **51** is coupled to the hybrid filter **11**. One of the selection terminals of the switch **51** is coupled to the filter **17**, and the other of the selection terminals of the switch **51** is coupled to the filter **18**. Specifically, the switch **51** is a TDD switch for selectively connecting the hybrid filter **11** to the low-noise amplifier **21** or the power amplifier **31**. The switch **51** is implemented by, for example, an SPDT switching circuit.

(59) The filters **15** and **16** may be implemented by one TDD filter and coupled between the hybrid filter **10** and the switch **50**. The filters **17** and **18** may be implemented as one TDD filter and coupled between the hybrid filter **11** and the switch **51**. The filters **15** to **18** and the matching circuits **45** to **48** are not necessarily included.

(60) The filters **15** and **16** may be implemented as one duplexer for providing transfer in accordance with a frequency division duplex (FDD) method. In this case, the switch **50** is unnecessary. The filters **17** and **18** may be implemented as one duplexer for providing transfer in accordance with an FDD method. In this case, the switch **51** is unnecessary.

(61) The low-noise amplifiers **20** and **21** and the switches **50** and **51** may be formed in one semiconductor integrated circuit (IC). The semiconductor IC may be implemented by, for example, a complementary metal-oxide semiconductor (CMOS) circuit. Specifically, the semiconductor IC is produced by a silicon on insulator (SOI) process. In this manner, the semiconductor ICs can be inexpensively manufactured. The semiconductor IC may be made of at least any of GaAs, SiGe, and GaN. This enables output of radio-frequency signals with high amplification performance and low-noise performance.

(62) With the circuit configuration described above, the radio-frequency module **1** is operable to solely transfer a transmit signal or receive signal in the first communication band and also operable to solely transfer a transmit signal or receive signal in the second communication band. The radio-frequency module **1** is also operable to simultaneously transfer a transmit signal or receive signal in the first communication band and a transmit signal or receive signal in the second communication band.

(63) It is sufficient that the radio-frequency module **1** according to the present embodiment includes, of the circuit components and circuit elements illustrated in FIG. **1**, at least the multiplexer **90**, the low-noise amplifiers **20** and **21**, and the matching circuits **45** and **47**. Alternatively, it is sufficient that the radio-frequency module **1** according to the present embodiment includes, of the circuit components and circuit elements illustrated in FIG. **1**, at least the multiplexer **90** and the switches **50** and **51**.

(64) Because the hybrid filter described above is formed by an acoustic wave resonator-inductor-capacitor combination, the hybrid filter includes many components. As a result, a problem arises in which the hybrid filter is relatively large in size. If the mounting density is increased for the purpose of reducing the size, another problem arises in which unnecessary coupling between components occurs, and the bandpass characteristic of the hybrid filter deteriorates.

(65) In this respect, the following describes configurations of the hybrid filter **10** of a smaller size in which degradation of the bandpass characteristic is suppressed, and the multiplexer **90** and the radio-frequency module **1**.

(66) [2. Arrangement of Circuit Elements of Radio-Frequency Module **1A** According to First Example]

(67) FIG. **3A** provides schematic plan views of a configuration of a radio-frequency module **1A** according to a first example. FIG. **3B** is a schematic sectional view of the configuration of the radio-frequency module **1A** according to the first example, more specifically, a sectional view taken along line III-III in FIG. **3A**. In FIG. **3A**, (a) illustrates an arrangement of circuit components when, of major surfaces **80a** and **80b** of a module substrate **80** that are opposite to each other, the major surface **80a** is viewed from the front side in the positive direction of the z axis. In FIG. **3A**, (b) provides a cutaway view of the arrangement of circuit components when the major surface **80b** is viewed from the front side in the positive direction of the z axis. In FIG. **3A**, for ease of understanding of the positional relationship among the circuit components, the individual circuit components are assigned corresponding symbols representing the function of each circuit component, although the symbols are not displayed in the radio-frequency module **1A** in actual applications.

(68) The radio-frequency module **1A** according to the first example specifically presents an arrangement of the circuit elements of the radio-frequency module **1** according to the embodiment.

(69) As illustrated in FIGS. **3A** and **3B**, the hybrid filter **10** according to this example includes the module substrate **80** as well as the acoustic wave resonator elements, inductors, and capacitor. In addition to the circuit configuration illustrated in FIG. **1**, the radio-frequency module **1A** according to this example further includes the module substrate **80**, resin members **81** and **82**, external connection terminals **150**, and a metal shield layer **85**.

(70) The module substrate **80** is an example of a substrate. The module substrate **80** has the major surface **80a** and the major surface **80b** that are opposite to each other. On the module substrate **80**,

the circuit components constituting the radio-frequency module **1A** are mounted. As the module substrate **80**, for example, a low temperature co-fired ceramics (LTCC) substrate having a layered structure composed of a plurality of dielectric layers, a high temperature co-fired ceramics (HTCC) substrate, a component-embedded substrate, a substrate including a redistribution layer (RDL), or a printed-circuit board is used.

(71) In this example, the major surface **80a** corresponds to a first major surface, and the major surface **80b** corresponds to a second major surface.

(72) It is desirable that the module substrate **80** have a multilayer structure including a plurality of dielectric layers stacked, and a ground electrode pattern be formed at least one of the plurality of dielectric layers. This improves the electromagnetic field blocking capability of the module substrate **80**.

(73) As illustrated in (b) of FIG. 3A, the antenna connection terminal **100**, the transmit input terminals **130** and **131**, and the receive output terminals **120** and **121** may be formed at the major surface **80b**.

(74) The resin member **81** is disposed at the major surface **80a** to cover a portion of the circuit components constituting the radio-frequency module **1A** and the major surface **80a**. The resin member **82** is disposed at the major surface **80b** to cover a portion of the circuit components constituting the radio-frequency module **1A** and the major surface **80b**. The resin members **81** and **82** have the function of maintaining the security of the properties of the circuit components constituting the radio-frequency module **1A** such as mechanical strength and moisture resistance.

(75) The metal shield layer **85** covers the surface of the resin member **81**. The metal shield layer **85** is set at a ground potential. The metal shield layer **85** is, for example, a metal thin film formed by sputtering.

(76) The resin members **81** and **82** and the metal shield layer **85** are non-essential constituent elements in the radio-frequency module **1** according to the present embodiment.

(77) In this example, the matching circuits **45**, **46**, **47**, and **48** each include an inductor.

(78) The inductors **L1**, **L2**, **L3**, **L4**, **L5**, and **L6**, which constitute the hybrid filter **10** or **11**, are an example of a first inductor that is disposed at the module substrate **80**. The capacitors **C3** and **C4**, which constitute the hybrid filter **10** or **11**, are an example of a first capacitor that is disposed at the module substrate **80**.

(79) Although not illustrated in FIG. 3A, interconnections connecting the circuit components illustrated in FIG. 1 are formed inside the module substrate **80** and at the major surfaces **80a** and **80b**. The interconnections may be bonding wires with ends joined to the major surfaces **80a** and **80b** and any of the circuit components constituting the radio-frequency module **1A**, or connectors, electrodes, or wires that are formed on the surface of the circuit components constituting the radio-frequency module **1A**.

(80) As illustrated in FIG. 3A, in the radio-frequency module **1A** according to this example, the acoustic wave resonators **A1** and **A2** and the inductors **L1**, **L2**, **L3**, **L4**, **L5**, and **L6** are an example of a first circuit element, disposed at the major surface **80a**; the capacitors **C3** and **C4** are an example of a second circuit element, disposed at the major surface **80b**.

(81) In the configuration described above, the acoustic wave resonator **A1** and the inductors **L1** to **L3**, and the capacitor **C3**, which constitute the hybrid filter **10**, are disposed across the module substrate **80**, separately on both sides of the module substrate. This arrangement reduces the size of the hybrid filter **10**. Further, unnecessary coupling between the acoustic wave resonator **A1** and the inductors **L1** to **L3**, and the capacitor **C3** is hindered. As a result, it is possible to provide the hybrid filter **10** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(82) Similarly, the acoustic wave resonator **A2** and the inductors **L4** to **L6**, and the capacitor **C4**, which constitute the hybrid filter **11**, are disposed across the module substrate **80**, separately on both sides of the module substrate. As a result, it is possible to provide the hybrid filter **11** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(83) Furthermore, it is possible to provide the multiplexer **90** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(84) It is sufficient that in either the hybrid filter **10** or **11**, the first circuit element and the second circuit element are disposed across the module substrate **80**, separately on both sides of the module substrate. As a result, it is possible to provide the hybrid filter **10** or **11** in which degradation of the bandpass characteristic is suppressed.

(85) In the radio-frequency module **1A** according to this example, the power amplifiers **30** and **31**, the filters **15** to **18**, and the matching circuits **45** to **48** are disposed at the major surface **80a**, whereas the low-noise amplifiers **20** and **21** and the switches **50** and **51** are disposed at the major surface **80b**.

(86) In the radio-frequency module **1A** according to this example, a plurality of external connection terminals **150** are disposed at the major surface **80b**. The radio-frequency module **1A** exchanges electrical signals with an external substrate provided on the front side in the negative direction of the z axis with respect to the radio-frequency module **1A**, through the plurality of external connection terminals **150**. Some of the external connection terminals **150** are set at the ground potential of the external substrate. Of the major surfaces **80a** and **80b**, at the major surface **80b** facing the external substrate, circuit components that cannot be easily formed as low-profile structures are not disposed, but the low-noise amplifiers **20** and **21** and the switches **50** and **51**, which can be easily formed as low-profile structures, are disposed at the major surface **80b** facing the external substrate.

(87) The external connection terminals **150** may be columnar electrodes extended in the Z-axis direction through the resin member **82** as illustrated in FIGS. **3A** and **3B**. Alternatively, the external connection terminals **150** may be bump electrodes formed on the major surface **80b**. In this case, the resin member **82** on the major surface **80b** is not necessarily provided.

(88) In the radio-frequency module **1A** according to this example, the capacitors **C3** and **C4** and the switches **50** and **51** are included in one semiconductor IC **71**. As a result, the capacitors **C3** and **C4** and the switches **50** and **51** have small-size low-profile structures.

(89) The capacitors **C3** and **C4** may be, for example, integrated passive devices (IPDs) disposed inside a Si substrate or at a surface of the Si substrate in an integrated manner. As a result, because the capacitors **C3** and **C4** are implemented by IPDs, which can be formed as low-profile structures, it is possible to reduce the height of the portion on the major surface **80b** side.

(90) In the radio-frequency module **1A** according to this example, the power amplifiers **30** and **31** are disposed at the major surface **80a**, whereas the low-noise amplifiers **20** and **21** are disposed at the major surface **80b**. In this arrangement, amplifiers for transmission and amplifiers for reception are disposed across the module substrate **80**, and thus, the isolation between the transmit and receive sides in the radio-frequency module **1A** is improved.

(91) In the radio-frequency module **1A** according to this example, the low-noise amplifiers **20** and **21** are included in one semiconductor IC **72**. As a result, the low-noise amplifiers **20** and **21** have small-size low-profile structures.

(92) The amount of heat generation of the power amplifiers **30** and **31** is relatively large among the circuit components included in the radio-frequency module **1A**. To enhance the heat dissipation capability of the radio-frequency module **1A**, it is important to dissipate heat generated in the power amplifiers **30** and **31** along heat dissipation paths of a low thermal resistance to an external substrate. If the power amplifiers **30** and **31** are arranged on the major surface **80b**, electrode wires coupled to the power amplifiers **30** and **31** are disposed at the major surface **80b**. As a result, the heat dissipation paths include heat dissipation paths routed via only planar interconnection patterns (along an xy plane) on the major surface **80b**. The planar interconnection patterns are formed by a metal thin film, and thus, the thermal resistance of the planar interconnection patterns is relatively high. As a result, if the power amplifiers **30** and **31** are disposed at the major surface **80b**, the heat dissipation capability is degraded.

(93) In this respect, in the radio-frequency module **1A** according to this example, the power amplifiers **30** and **31** can be coupled to the external connection terminals **150** by via-conductors for heat dissipation coupled to ground electrodes of the power amplifiers **30** and **31**, extended from the major surface **80a** to the major surface **80b**. As such, the heat dissipation paths of the power amplifiers **30** and **31** do not include heat dissipation paths routed via only planar interconnection patterns along an xy plane, the thermal resistance of which is relatively high among the interconnections in the module substrate **80**. As a result, it is possible to provide the radio-frequency module **1A** of a smaller size with an improved capability of dissipating heat from the power amplifiers **30** and **31** to an external substrate.

(94) In the radio-frequency module **1A** according to this example, it is necessary that one of the acoustic wave resonator elements **P1** and **P2** and the inductors **L1**, **L2**, and **L3** is disposed at the major surface **80a**, and the capacitor **C3** is disposed at the major surface **80b**; the other circuit components may be disposed at either of the major surfaces **80a** and **80b**.

(95) In the radio-frequency module **1A** according to this example, either one of the hybrid filters **10** and **11** may have a pass band including at least one of the followings: Fourth Generation Long-Term Evolution (4G-LTE) Band 42 (3400-3600 MHz), Band 43 (3600-3800 MHz), Band 48 (3550-3700 MHz), Band **49** (3550-3700 MHz), Fifth Generation New Radio (5G-NR) n77 (3300-4200 MHz), n78 (3300-3800 MHz), n79 (4400-5000 MHz), n46 (5150-5925 MHz), n96 (5925-7125 MHz), n97, Wireless Local Area Network (WLAN) (5150-5350 MHz), WLAN (5470-5850 MHz), or WLAN (5925-7125 MHz).

(96) As a result, the hybrid filter **10** or **11** having wide-band and steep filter characteristics is usable for the above communication bands that are wide ranges close to adjacent bands.

(97) In the radio-frequency module **1A** according to this example, the pass band of the hybrid filter **10** may include 5G-NR n79, and the pass band of the hybrid filter **11** may include at least one selected from 4GLTE Band 42, Band 43, Band 48, Band 49, 5G-NR n77, and n78.

(98) [3. Arrangement of Circuit Elements of Radio-Frequency Module **1B** According to Second Example]

(99) FIG. **4** provides schematic plan views of a configuration of a radio-frequency module **1B** according to a second example. In FIG. **4**, (a) illustrates an arrangement of circuit components when, of major surfaces **80a** and **80b** of a module substrate **80** that are opposite to each other, the major surface **80a** is viewed from the front side in the positive direction of the z axis. In FIG. **4**, (b) provides a cutaway view of the arrangement of circuit components when the major surface **80b** is viewed from the front side in the positive direction of the z axis. In FIG. **4**, for ease of understanding of the positional relationship among the circuit components, the individual circuit components are assigned corresponding symbols representing the function of each circuit component, although the symbols are not displayed in the radio-frequency module **1B** in actual applications.

(100) The radio-frequency module **1B** according to the second example specifically presents an arrangement of the circuit elements of the radio-frequency module **1** according to the embodiment.

(101) The radio-frequency module **1B** according to this example differs from the radio-frequency module **1A** according to the first example in the arrangement of the circuit components constituting the hybrid filters **10** and **11**. The following describes the radio-frequency module **1B** according to this example with a main focus on configurations different from the radio-frequency module **1A** according to the first example, and descriptions of the same configurations as the radio-frequency module **1A** according to the first practical example are not repeated.

(102) As illustrated in FIG. **4**, in the radio-frequency module **1B** according to this example, the acoustic wave resonators **A1** and **A2**, the inductors **L1**, **L2**, **L5**, and **L6**, and the capacitors **C3** and **C4** are an example of the first circuit element, disposed at the major surface **80a**; the inductors **L3** and **L4** are an example of the second circuit element, disposed at the major surface **80b**.

(103) In the configuration described above, the acoustic wave resonator **A1**, the inductors **L1** and

L2, and the capacitor C3, and the inductor L3, which constitute the hybrid filter 10, are disposed across the module substrate 80, separately on both sides of the module substrate. This arrangement reduces the size of the hybrid filter 10. Further, unnecessary coupling between the acoustic wave resonator A1, the inductors L1 and L2, and the capacitor C3, and the inductor L3 is hindered. As a result, it is possible to provide the hybrid filter 10 of a smaller size in which degradation of the bandpass characteristic is suppressed.

(104) Similarly, the acoustic wave resonator A2, the inductors L5 and L6, and the capacitor C4, and the inductor L4, which constitute the hybrid filter 11, are disposed across the module substrate 80, separately on both sides of the module substrate. As a result, it is possible to provide the hybrid filter 11 of a smaller size in which degradation of the bandpass characteristic is suppressed.

(105) Furthermore, it is possible to provide the multiplexer 90 of a smaller size in which degradation of the bandpass characteristic is suppressed.

(106) In the radio-frequency module 1B according to this example, the switches 50 and 51 are included in the one semiconductor IC 71. As a result, the switches 50 and 51 have small-size low-profile structures.

(107) The inductors L3 and L4 may be, for example, IPDs disposed inside a Si substrate or at a surface of the Si substrate in an integrated manner. As a result, because the inductors L3 and L4 are implemented by IPDs, which can be formed as low-profile structures, it is possible to reduce the height of the portion on the major surface 80b side.

(108) In the radio-frequency module 1B according to this example, it is necessary that one of the acoustic wave resonator elements P1 and P2 and the capacitor C3 is disposed at the major surface 80a, and one of the inductors L1, L2, and L3 is disposed at the major surface 80b; the other circuit components may be disposed at either of the major surfaces 80a and 80b.

(109) [4. Arrangement of Circuit Elements of Radio-Frequency Module 1C According to Third Example]

(110) FIG. 5 provides schematic plan views of a configuration of a radio-frequency module 1C according to a third example. In FIG. 5, (a) illustrates an arrangement of circuit components when, of major surfaces 80a and 80b of a module substrate 80 that are opposite to each other, the major surface 80a is viewed from the front side in the positive direction of the z axis. In FIG. 5, (b) provides a cutaway view of the arrangement of circuit components when the major surface 80b is viewed from the front side in the positive direction of the z axis. In FIG. 5, for ease of understanding of the positional relationship among the circuit components, the individual circuit components are assigned corresponding symbols representing the function of each circuit component, although the symbols are not displayed in the radio-frequency module 1C in actual applications.

(111) The radio-frequency module 1C according to the third example specifically presents an arrangement of the circuit elements of the radio-frequency module 1 according to the embodiment.

(112) The radio-frequency module 1C according to this example differs from the radio-frequency module 1A according to the first example in the arrangement of the circuit components constituting the hybrid filters 10 and 11. The following describes the radio-frequency module 1C according to this example with a main focus on configurations different from the radio-frequency module 1A according to the first example, and descriptions of the same configurations as the radio-frequency module 1A according to the first practical example are not repeated.

(113) As illustrated in FIG. 5, in the radio-frequency module 1C according to this example, the acoustic wave resonator elements P1 and P5, the inductors L1, L2, L3, L4, L5, and L6, and the capacitors C3 and C4 are an example of the first circuit element, disposed at the major surface 80a; the acoustic wave resonator elements P2 and P6 are an example of the second circuit element, disposed at the major surface 80b.

(114) In the configuration described above, the acoustic wave resonator element P1, the inductors L1 to L3, and the capacitor C3, and the acoustic wave resonator element P2, which constitute the

hybrid filter **10**, are disposed across the module substrate **80**, separately on both sides of the module substrate. This arrangement reduces the size of the hybrid filter **10**. Further, unnecessary coupling between the acoustic wave resonator element **P1**, the inductors **L1** to **L3**, and the capacitor **C3**, and the acoustic wave resonator element **P2** is hindered. As a result, it is possible to provide the hybrid filter **10** of a smaller size in which degradation of the bandpass characteristic is suppressed. (115) Similarly, the acoustic wave resonator element **P5**, the inductors **L4** to **L6**, and the capacitor **C4**, and the acoustic wave resonator element **P6**, which constitute the hybrid filter **11**, are disposed across the module substrate **80**, separately on both sides of the module substrate. As a result, it is possible to provide the hybrid filter **11** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(116) Furthermore, it is possible to provide the multiplexer **90** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(117) In the radio-frequency module **1C** according to this example, the switches **50** and **51** are included in the one semiconductor IC **71**. As a result, the switches **50** and **51** have small-size low-profile structures.

(118) In the radio-frequency module **1C** according to this example, it is necessary that the acoustic wave resonator element **P1** is disposed at the major surface **80a**, and the acoustic wave resonator element **P2** is disposed at the major surface **80b**; the other circuit components may be disposed at either of the major surfaces **80a** and **80b**.

(119) [5. Arrangement of Circuit Elements of Radio-Frequency Module **1D** According to Fourth Example]

(120) FIG. **6** is a schematic sectional view of a configuration of a radio-frequency module **1D** according to a fourth example. The radio-frequency module **1D** according to the fourth example specifically presents an arrangement of the circuit elements of the radio-frequency module **1** according to the embodiment.

(121) The radio-frequency module **1D** according to this example differs from the radio-frequency module **1A** according to the first example in the arrangement of the circuit components constituting the hybrid filter **10**. The following describes the radio-frequency module **1D** according to this example with a main focus on configurations different from the radio-frequency module **1A** according to the first example, and descriptions of the same configurations as the radio-frequency module **1A** according to the first practical example are not repeated.

(122) As illustrated in FIG. **6**, in the radio-frequency module **1D** according to this example, the acoustic wave resonators **A1** and **A2** and the inductors **L1**, **L2**, **L4**, **L5**, and **L6** are an example of the first circuit element, disposed at the major surface **80a**; the capacitors **C3** and **C4** are an example of the second circuit element, disposed at the major surface **80b**.

(123) In the configuration described above, the acoustic wave resonator **A1** and the inductors **L1** and **L2**, and the capacitor **C3**, which constitute the hybrid filter **10**, are disposed across the module substrate **80**, separately on both sides of the module substrate. This arrangement reduces the size of the hybrid filter **10**. Further, unnecessary coupling between the acoustic wave resonator **A1** and the inductors **L1** and **L2**, and the capacitor **C3** is hindered. As a result, it is possible to provide the hybrid filter **10** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(124) Similarly, the acoustic wave resonator **A2** and the inductors **L4** to **L6**, and the capacitor **C4**, which constitute the hybrid filter **11**, are disposed across the module substrate **80**, separately on both sides of the module substrate. As a result, it is possible to provide the hybrid filter **11** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(125) It is sufficient that in either the hybrid filter **10** or **11**, the first circuit element and the second circuit element are disposed across the module substrate **80**, separately on both sides of the module substrate. As a result, it is possible to provide the hybrid filter **10** or **11** in which degradation of the bandpass characteristic is suppressed.

(126) As illustrated in FIG. **6**, the inductor **L3** is formed inside the module substrate **80**.

(127) This arrangement further reduces the size of the hybrid filter **10**. At least one of the inductors **L1** to **L3** and the capacitor **C3**, which constitute the hybrid filter **10**, may be formed inside the module substrate **80**. At least one of the inductors **L4** to **L6** and the capacitor **C4**, which constitute the hybrid filter **11**, may be formed inside the module substrate **80**.

(128) [6. Arrangement of Circuit Elements of Radio-Frequency Modules According to Modifications]

(129) In the radio-frequency modules according to the first to fourth examples, the matching circuit **45** coupled to the input terminal of the low-noise amplifier **20** is disposed at the major surface **80a**. In this case, when magnetic field coupling between the inductors **L1** to **L3** of the hybrid filter **10** and the inductor (a second inductor) of the matching circuit **45** is relatively strong, it is assumed that a transmit signal outputted from the power amplifier **30**, a harmonic wave of the transmit signal, or an intermodulation distortion can enter the receive path including the low-noise amplifier **20** through the magnetic field coupling, and the receive sensitivity can thus be degraded. In this respect, in first to third modifications, configurations are presented in which degradation of the receive sensitivity is suppressed.

(130) FIG. 7A is a schematic partial plan view of a configuration of a radio-frequency module according to the first modification. This drawing illustrates an arrangement of circuit components in a partial area of the major surface **80a** of the module substrate **80**.

(131) As illustrated in FIG. 7A, when the module substrate **80** is viewed in plan view, the acoustic wave resonator **A1** is disposed between the inductors **L1** to **L3** and the inductor of the matching circuit **45**. With this arrangement, the acoustic wave resonator **A1** hinders coupling between the magnetic flux generated around the inductors **L1** to **L3** and the magnetic flux generated around the inductor of the matching circuit **45**. As such, it is possible to reduce the likelihood of magnetic field coupling between the inductors **L1** to **L3** and the inductor of the matching circuit **45**.

(132) The circuit component disposed between the inductors **L1** to **L3** and the inductor of the matching circuit **45** is not limited to the acoustic wave resonator **A1**; any of the other circuit components may be disposed between the inductors **L1** to **L3** and the inductor of the matching circuit **45**.

(133) FIG. 7B is a schematic partial plan view of a configuration of a radio-frequency module according to the second modification. This drawing illustrates an arrangement of circuit components in a partial area of the major surface **80a** of the module substrate **80**.

(134) As illustrated in FIG. 7B, when the module substrate **80** is viewed in plan view, the winding axis direction (x-axis direction) of the inductors **L1** to **L3** and the winding axis direction (y-axis direction) of the inductor of the matching circuit **45** cross each other at the major surface **80a**. In other words, the winding axes of the inductors **L1** to **L3** are not parallel with the winding axis of the inductor of the matching circuit **45**. This arrangement reduces the likelihood of magnetic field coupling between the inductors **L1** to **L3** and the inductor of the matching circuit **45**.

(135) FIG. 7C is a schematic partial plan view of a configuration of a radio-frequency module according to the third modification. This drawing illustrates an arrangement of circuit components in a partial area of the major surface **80a** of the module substrate **80**.

(136) As illustrated in FIG. 7C, the winding axis direction (x-axis direction) of the inductors **L1** to **L3** and the winding axis direction (z-axis direction) of the inductor of the matching circuit **45** cross each other. In other words, the winding axes of the inductors **L1** to **L3** are not parallel with the winding axis of the inductor of the matching circuit **45**. This arrangement reduces the likelihood of magnetic field coupling between the inductors **L1** to **L3** and the inductor of the matching circuit **45**.

(137) Although not illustrated in the drawings, it may be possible that the winding axes of the inductors **L1** to **L3** and the winding axis of the inductor of the matching circuit **45** are noncollinear. Also in this case, as compared to the case in which the winding axes of the inductors **L1** to **L3** and the winding axis of the inductor of the matching circuit **45** are collinear, the likelihood of magnetic field coupling between the inductors **L1** to **L3** and the inductor of the matching circuit **45** is

reduced.

(138) [7. Arrangement of Circuit Elements of Radio-Frequency Module **1E** According to Fifth Example]

(139) In the radio-frequency module **1** and the communication device **5**, the hybrid filters **10** and **11** operate as TDD filters operable for both transmission and reception. Specifically, the hybrid filter **10** filters a receive signal received by the antenna **2** and outputs the receive signal to the receive path including the switch **50**, the filter **15**, the matching circuit **45**, the low-noise amplifier **20**, and the receive output terminal **120**; the hybrid filter **10** also filters a transmit signal transferred through the transmit path including the transmit input terminal **130**, the power amplifier **30**, the matching circuit **46**, the filter **16**, and the switch **50** and outputs the transmit signal to the antenna **2**. The hybrid filter **11** filters a receive signal received by the antenna **2** and outputs the receive signal to the receive path including the switch **51**, the filter **17**, the matching circuit **47**, the low-noise amplifier **21**, and the receive output terminal **121**; the hybrid filter **11** also filters a transmit signal transferred through the transmit path including the transmit input terminal **131**, the power amplifier **31**, the matching circuit **48**, the filter **18**, and the switch **51** and outputs the transmit signal to the antenna **2**.

(140) In contrast, a radio-frequency module **1E** according to this example includes a radio-frequency circuit **61** including the hybrid filters **10** and **11** that are operable for both transmission and reception, and additionally, a radio-frequency circuit **62** including hybrid filters **12** and **13** that are configured especially for reception. The following describes the radio-frequency module **1E** and a communication device **5E** according to this example with a main focus on configurations different from the radio-frequency module **1** and the communication device **5**.

(141) FIG. **8** is a circuit configuration diagram of the radio-frequency module **1E** and the communication device **5E** according to a fifth example. As illustrated in the drawing, the communication device **5E** includes the radio-frequency module **1E**, antennas **2A** and **2B**, the RFIC **3**, and the BBIC **4**.

(142) The antenna **2A** is coupled to an antenna connection terminal **200** of the radio-frequency module **1E**. The antenna **2A** emits a radio-frequency signal outputted from the radio-frequency module **1E**. The antenna **2A** also receives a radio-frequency signal from outside and outputs the radio-frequency signal to the radio-frequency module **1E**.

(143) The antenna **2B** is coupled to an antenna connection terminal **210** of the radio-frequency module **1E**. The antenna **2B** emits a radio-frequency signal outputted from the radio-frequency module **1E**. The antenna **2B** also receives a radio-frequency signal from outside and outputs the radio-frequency signal to the radio-frequency module **1E**.

(144) The following describes a configuration of the radio-frequency module **1E**. The radio-frequency module **1E** includes the antenna connection terminals **200** and **210**, a switch **60**, and the radio-frequency circuits **61** and **62**.

(145) The switch **60** includes common terminals **60a** and **60b** and selection terminals **60c**, **60d**, **60e**, and **60f**. The switch **60** controls connection and disconnection between the common terminal **60a** and at least one of the selection terminals **60c** to **60f** and connection and disconnection between the common terminal **60b** and at least one of the selection terminals **60c** to **60f**. The common terminal **60a** is coupled to the antenna connection terminal **200**. The common terminal **60b** is coupled to the antenna connection terminal **210**. The selection terminal **60c** is coupled to the hybrid filter **10**. The selection terminal **60d** is coupled to the hybrid filter **11**. The selection terminal **60e** is coupled to the hybrid filter **12**. The selection terminal **60f** is coupled to the hybrid filter **13**.

(146) With the connection configuration of the switch **60** described above, in the communication device **5E**, it is possible to couple the antenna **2A** to at least one of the hybrid filters **10** to **13** and couple the antenna **2B** to at least one of the hybrid filters **10** to **13**.

(147) The radio-frequency circuit **61** includes the receive output terminals **120** and **121**, the transmit input terminals **130** and **131**, the hybrid filters **10** and **11**, the switches **50** and **51**, the

matching circuits **45**, **46**, **47**, and **48**, the low-noise amplifiers **20** and **21**, and the power amplifiers **30** and **31**. The radio-frequency circuit **61** may include the filters **15** to **18**, which are included in the radio-frequency module **1**.

(148) The radio-frequency circuit **62** includes receive output terminals **122** and **123**, the hybrid filters **12** and **13**, matching circuits **43** and **44**, and low-noise amplifiers **22** and **23**.

(149) The hybrid filters **12** and **13** each includes one or more acoustic wave resonator elements, one or more inductors, and one or more capacitors. One terminal of the hybrid filter **12** is coupled to the selection terminal **60e**, and the other terminal is coupled to the low-noise amplifier **22** via the matching circuit **43**. One terminal of the hybrid filter **13** is coupled to the selection terminal **60f**, and the other terminal is coupled to the low-noise amplifier **23** via the matching circuit **44**. The hybrid filters **12** and **13** form a multiplexer **91**.

(150) The acoustic wave resonator elements included in the hybrid filters **12** and **13** are, for example, acoustic wave resonator elements using SAWs or acoustic wave resonator elements using BAWs.

(151) It is sufficient that the multiplexer **91** is configured to separate at least two different frequency ranges (communication bands) of radio-frequency signals. According to this, the number of filters forming the multiplexer **91** may be two or more.

(152) The specific circuit configuration of the hybrid filters **12** and **13** is exemplified by the circuit configurations illustrated in FIGS. 2A and 2B.

(153) The low-noise amplifier **22** amplifies receive signals in, for example, the first communication band with low noise and outputs the receive signals to the receive output terminal **122**. The low-noise amplifier **23** amplifies receive signals in, for example, the second communication band with low noise and outputs the receive signals to the receive output terminal **123**.

(154) The matching circuit **43** is coupled between the low-noise amplifier **22** and the hybrid filter **12**. The matching circuit **43** provides impedance matching between the low-noise amplifier **22** and the hybrid filter **12**. The matching circuit **44** is coupled between the low-noise amplifier **23** and the hybrid filter **13**. The matching circuit **44** provides impedance matching between the low-noise amplifier **23** and the hybrid filter **13**.

(155) The low-noise amplifiers **22** and **23** may be formed in one semiconductor IC. The semiconductor IC may be implemented by, for example, a CMOS circuit. Specifically, the semiconductor IC is produced by an SOI process. In this manner, the semiconductor ICs can be inexpensively manufactured. The semiconductor IC may be made of at least any of GaAs, SiGe, and GaN. This enables output of radio-frequency signals with high amplification performance and low-noise performance.

(156) The low-noise amplifiers **20**, **21**, **22**, and **23** may be formed in one semiconductor IC. The low-noise amplifiers **22** and **23** and the switch **60** may be formed in one semiconductor IC.

(157) With the circuit configuration described above, the radio-frequency module **1E** is operable to transfer at least one signal selected from the group consisting of a transmit signal and two kinds of receive signals in the first communication band and also operable to transfer at least one signal selected from the group consisting of a transmit signal and two kinds of receive signals in the second communication band. The radio-frequency module **1E** is also operable to simultaneously transfer at least two signals selected from the group consisting of a transmit signal and receive signals in the first communication band and a transmit signal and receive signals in the second communication band.

(158) The circuit elements constituting the radio-frequency module **1E** according to this example are disposed at the major surface **80a** or **80b** of the module substrate **80** or inside the module substrate **80**.

(159) The circuit elements constituting the radio-frequency circuit **61** are disposed, for example, in the same arrangement as FIGS. 3A and 3B. Specifically, the acoustic wave resonators **A1** and **A2** and the inductors **L1**, **L2**, **L3**, **L4**, **L5**, and **L6** are disposed at the major surface **80a**, and the

capacitors **C3** and **C4** are disposed at the major surface **80b**.

(160) The circuit elements constituting the radio-frequency circuit **62** may be disposed at the major surface **80a** or **80b** of the module substrate **80** or inside the module substrate **80**.

(161) In the configuration described above, the acoustic wave resonator **A1** and the inductors **L1** to **L3**, and the capacitor **C3**, which constitute the hybrid filter **10**, are disposed across the module substrate **80**, separately on both sides of the module substrate. Similarly, the acoustic wave resonator **A2**, the inductors **L4** to **L6**, and the capacitor **C4**, which constitute the hybrid filter **11**, are disposed across the module substrate **80**, separately on both sides of the module substrate. This arrangement reduces the size of the hybrid filters **10** and **11** and consequently reduces the size of the radio-frequency module **1E** and the size of the communication device **5E**. Furthermore, this arrangement hinders unnecessary coupling between the acoustic wave resonator **A1** and the inductors **L1** to **L3**, and the capacitor **C3** and also hinders unnecessary coupling between the acoustic wave resonator **A2** and the inductors **L4** to **L6**, and the capacitor **C4**. As a result, it is possible to provide the radio-frequency module **1E** and the communication device **5E** including the hybrid filters **10** and **11** in which degradation of the bandpass characteristic is suppressed.

(162) It is desirable that the switch **60** be disposed at the major surface **80b** of the module substrate **80**. It is also desirable that the switch **60** as well as the capacitors **C3** and **C4** be included in one semiconductor IC.

(163) The circuit elements constituting the radio-frequency circuit **61** may be disposed in the same arrangement as FIG. **4**, **5**, **6**, or any of **7A** to **7C**.

(164) [8. Effects]

(165) As described above, the hybrid filter **10** according to the first to fifth examples includes the module substrate **80** having the major surfaces **80a** and **80b** that are opposite to each other, the acoustic wave resonator elements **P1** and **P2** disposed at the module substrate **80**, the inductors **L1** to **L3** disposed at the module substrate **80**, and the capacitor **C3** disposed at the module substrate **80**. The pass band of the hybrid filter **10** is wider than the resonance band width of the acoustic wave resonator elements **P1** and **P2**. Any of the acoustic wave resonator elements **P1** and **P2**, the inductors **L1** to **L3**, and the capacitor **C3** is the first circuit element, and the first circuit element is disposed at the major surface **80a**. Any of the acoustic wave resonator elements **P1** and **P2**, the inductors **L1** to **L3**, and the capacitor **C3** is the second circuit element, and the second circuit element is disposed at the major surface **80b**.

(166) In this arrangement, the first circuit element and the second circuit element, which constitute the hybrid filter **10**, are disposed across the module substrate **80**, separately on both sides of the module substrate **80**. This arrangement reduces the size of the hybrid filter **10**. Furthermore, unnecessary coupling between the first circuit element and the second circuit element is hindered. As a result, it is possible to provide the hybrid filter **10** of a smaller size in which degradation of the bandpass characteristic is suppressed.

(167) In the hybrid filter **10** according to the first, second, fourth, and fifth examples, the first circuit element may be the acoustic wave resonator elements **P1** and **P2**, and the second circuit element may be the inductors **L1** to **L3** or the capacitor **C3**.

(168) In the hybrid filter **10** according to the first, second, fourth, and fifth examples, the second circuit element may be one of the inductors **L1** to **L3** and the capacitor **C3**, and the first circuit element may be the other of the inductors **L1** to **L3** and the capacitor **C3**.

(169) In the hybrid filter **10** according to the first, fourth, and fifth examples, the second circuit element may be the capacitor **C3**.

(170) In the hybrid filter **10** according to the third and fifth examples, the first circuit element may be the acoustic wave resonator element **P1**, and the second circuit element may be the acoustic wave resonator element **P2**.

(171) In the hybrid filter **10** according to the fourth and fifth examples, the inductors **L1** to **L3** or the capacitor **C3** may be formed inside the module substrate **80**.

(172) This arrangement further reduces the size of the hybrid filter **10**.

(173) In the first to fifth examples, the pass band of the hybrid filter **10** may include at least one selected from 4G-LTE Band 42, Band 43, Band 48, Band 49, 5G-NR n77, n78, n79, n46, n96, n97, and WLAN.

(174) As a result, the hybrid filter **10** having wide-band and steep filter characteristics is usable for the above communication bands that are wide ranges close to adjacent bands.

(175) The multiplexer **90** according to the first to fifth examples includes the hybrid filters **10** and **11**.

(176) In the multiplexer **90** according to the first to fifth examples, the pass band of the hybrid filter **10** may include 5G-NR n79, and the pass band of the hybrid filter **11** may include at least one selected from 4G-LTE Band 42, Band 43, Band 48, Band 49, 5G-NR n77, and n78.

(177) The radio-frequency module according to the first to fifth examples includes the hybrid filter **10**, the low-noise amplifier **20**, the power amplifier **30**, and the switch **50** for selectively connecting the hybrid filter **10** to the low-noise amplifier **20** or the power amplifier **30**.

(178) The radio-frequency module according to the first to fifth examples may include the external connection terminals **150** disposed at the major surface **80b**.

(179) In the radio-frequency module **1A** according to the first example, the switch **50** may be disposed at the major surface **80b**, and the capacitor **C3** disposed at the major surface **80b** and the switch **50** may be included in the one semiconductor IC **71**.

(180) As a result, the radio-frequency module **1A** has a small-size low-profile structure.

(181) In the radio-frequency module **1B** according to the second example, the inductor **L3** disposed at the major surface **80b** may be an integrated passive device.

(182) As a result, the radio-frequency module **1B** has a small-size low-profile structure.

(183) In the radio-frequency module according to the first to fifth examples, the low-noise amplifier **20** may be disposed at the major surface **80b**; the radio-frequency module according to the first to fifth examples may further include the inductor of the matching circuit **45** coupled to the input terminal of the low-noise amplifier **20**, disposed at the major surface **80a**.

(184) In the radio-frequency module according to the second and third modifications, it may be possible that the winding axes of the coils forming the inductors **L1** to **L3** are not parallel with the winding axis of the coil forming the inductor of the matching circuit **45**.

(185) This arrangement reduces the likelihood of magnetic field coupling between the inductors **L1** to **L3** and the inductor of the matching circuit **45**. As a result, degradation of the receive sensitivity of the radio-frequency module is suppressed.

(186) In the radio-frequency module according to the first to third modifications, it may be possible that the winding axes of the coils forming the inductors **L1** to **L3** and the winding axis of the coil forming the inductor of the matching circuit **45** are noncollinear.

(187) With this arrangement, as compared to the case in which the winding axes of the inductors **L1** to **L3** and the winding axis of the inductor of the matching circuit **45** are collinear, the likelihood of magnetic field coupling between the inductors **L1** to **L3** and the inductor of the matching circuit **45** is reduced. As a result, degradation of the receive sensitivity of the radio-frequency module is suppressed.

(188) In the radio-frequency module according to the first modification, a circuit component may be disposed between the inductors **L1** to **L3** and the inductor of the matching circuit **45**.

(189) With this arrangement, the circuit component hinders coupling between the magnetic flux generated around the inductors **L1** to **L3** and the magnetic flux generated around the inductor of the matching circuit **45**. As such, it is possible to reduce the likelihood of magnetic field coupling between the inductors **L1** to **L3** and the inductor of the matching circuit **45**. As a result, degradation of the receive sensitivity of the radio-frequency module is suppressed.

(190) The radio-frequency module according to the first to fifth examples may further include the filter **15** coupled between the low-noise amplifier **20** and the hybrid filter **10**.

(191) In the radio-frequency module according to the first to fifth examples, the power amplifier **30** may be disposed at the major surface **80a**.

(192) In this arrangement, amplifiers for transmission and amplifiers for reception are disposed across the module substrate **80**, and thus, the isolation between the transmit and receive sides in the radio-frequency module is improved.

(193) The communication device **5** includes the RFIC **3** for processing a radio-frequency signal received by the antenna **2** and the radio-frequency module **1** for transferring the radio-frequency signal between the antenna **2** and the RFIC **3**.

(194) As a result, it is possible to provide a communication device including a hybrid filter of a smaller size in which degradation of filter characteristics is suppressed.

Other Embodiments

(195) The hybrid filter, multiplexer, radio-frequency module, and communication device according to the present disclosure has been described by using the embodiment, examples, and modifications, but the present disclosure is not limited to the embodiment, examples, and modifications described above. The present disclosure also embraces other embodiments implemented as any combination of the constituent elements of the embodiment, examples, and modifications, other modified examples obtained by making various modifications to the embodiment that occur to those skilled in the art without departing from the scope of the present disclosure, and various hardware devices including the hybrid filter, multiplexer, radio-frequency module, or communication device according to the present disclosure.

(196) For example, in the hybrid filter, multiplexer, radio-frequency module, and communication device according to the embodiment, examples, and modifications, a matching element such as an inductor or capacitor and a switching circuit may be coupled among the constituent elements. The inductor may include a wire inductor formed by a wire serving as an interconnection between constituent elements.

INDUSTRIAL APPLICABILITY

(197) The present disclosure can be used, as a hybrid filter, a multiplexer, a radio-frequency module and a communication device operable in multiband systems, in a wide variety of communication devices such as mobile phones.

REFERENCE SIGNS LIST

(198) **1, 1A, 1B, 1C, 1D, 1E** radio-frequency module **2, 2A, 2B** antenna **3** RF signal processing circuit (RFIC) **4** baseband signal processing circuit (BBIC) **5, 5E** communication device **10, 11, 12, 13** hybrid filter **15, 16, 17, 18** filter **20, 21, 22, 23** low-noise amplifier **30, 31** power amplifier **43, 44, 45, 46, 47, 48** matching circuit **50, 51, 60** switch **60a, 60b** common terminal **60c, 60d, 60e, 60f** selection terminal **61, 62** radio-frequency circuit **71, 72** semiconductor IC **80** module substrate **80a, 80b** major surface **81, 82** resin member **85** metal shield layer **90, 91** multiplexer **100, 200, 210** antenna connection terminal **101, 102, 103, 104, 105, 106** input-output terminal **120, 121, 122, 123** receive output terminal **130, 131** transmit input terminal **150** external connection terminal **A1, A2** acoustic wave resonator **C3, C4, C6, C7, C8, C9** capacitor **L1, L2, L3, L4, L5, L6, L7, L8** inductor **P1, P2, P5, P6** acoustic wave resonator element

Claims

1. A hybrid filter comprising: a substrate having a first major surface and a second major surface that are opposite to each other; one or more acoustic wave resonator elements disposed at the substrate; one or more first inductors disposed at the substrate; and one or more first capacitors disposed at the substrate, wherein a pass band width of the hybrid filter is wider than a resonance band width of the one or more acoustic wave resonator elements, one of the one or more acoustic wave resonator elements, the one or more first inductors, and the one or more first capacitors is a first circuit element, and the first circuit element is disposed at the first major surface, and another

of the one or more acoustic wave resonator elements, the one or more first inductors, and the one or more first capacitors is a second circuit element, and the second circuit element is disposed at the second major surface, wherein one of the one or more first inductors and the one or more first capacitors is formed inside the substrate.

2. The hybrid filter according to claim 1, wherein the first circuit element is one of the one or more acoustic wave resonator elements, and the second circuit element is one of the one or more first inductors and the one or more first capacitors.

3. The hybrid filter according to claim 1, wherein the second circuit element is one selected from one of the one or more first inductors and one of the one or more first capacitors, and the first circuit element is another one selected from one of the one or more first inductors and one of the one or more first capacitors.

4. The hybrid filter according to claim 2, wherein the second circuit element is one of the one or more first capacitors.

5. The hybrid filter according to claim 1, wherein the first circuit element is one of the one or more acoustic wave resonator elements, and the second circuit element is another of the one or more acoustic wave resonator elements.

6. The hybrid filter according to claim 1, wherein a pass band of the hybrid filter includes at least one selected from Fourth Generation Long-Term Evolution (4G-LTE) Band 42, Band 43, Band 48, Band 49, Fifth Generation New Radio (5G-NR) n77, n78, n79, n46, n96, n97, Wireless Local Area Network (WLAN) (5150-5350 MHz), WLAN (5470-5850 MHz), and WLAN (5925-7125 MHz).

7. A multiplexer comprising: a common terminal; a first hybrid filter that is the hybrid filter according to claim 1, the first hybrid filter being coupled to the common terminal; and a second hybrid filter that is the hybrid filter according to claim 1, the second hybrid filter being coupled to the common terminal.

8. The multiplexer according to claim 7, wherein a pass band of the first hybrid filter includes 5G-NR n79, and a pass band of the second hybrid filter includes at least one selected from 4G-LTE Band 42, Band 43, Band 48, Band 49, 5G-NR n77, and n78.

9. A radio-frequency module comprising: a hybrid filter including a substrate having a first major surface and a second major surface that are opposite to each other, one or more acoustic wave resonator elements disposed at the substrate, one or more first inductors disposed at the substrate, and one or more first capacitors disposed at the substrate, wherein a pass band width of the hybrid filter is wider than a resonance band width of the one or more acoustic wave resonator elements, one of the one or more acoustic wave resonator elements, the one or more first inductors, and the one or more first capacitors is a first circuit element, and the first circuit element is disposed at the first major surface, and another of the one or more acoustic wave resonator elements, the one or more first inductors, and the one or more first capacitors is a second circuit element, and the second circuit element is disposed at the second major surface, wherein one of the one or more first inductors and the one or more first capacitors is formed inside the substrate; a low-noise amplifier; a power amplifier; and a switch configured to selectively connect the hybrid filter to the low-noise amplifier or the power amplifier.

10. The radio-frequency module according to claim 9, further comprising: a plurality of external connection terminals disposed at the second major surface.

11. The radio-frequency module according to claim 10, wherein the switch is disposed at the second major surface, and the second circuit element disposed at the second major surface and the switch are included in one semiconductor integrated circuit (IC).

12. The radio-frequency module according to claim 10, wherein the second circuit element is an integrated passive device.

13. The radio-frequency module according to claim 10, wherein the low-noise amplifier is disposed at the second major surface, the radio-frequency module further comprising: a second inductor coupled to an input terminal of the low-noise amplifier, the second inductor being disposed at the

first major surface.

14. The radio-frequency module according to claim 13, wherein a winding axis of a coil forming one of the one or more first inductors and a winding axis of a coil forming the second inductor are not parallel with each other.

15. The radio-frequency module according to claim 13, wherein a winding axis of a coil forming one of the one or more first inductors and a winding axis of a coil forming the second inductor are noncollinear.

16. The radio-frequency module according to claim 13, wherein a circuit component is disposed between one of the one or more first inductors, and the second inductor.

17. The radio-frequency module according to claim 13, further comprising: an LC filter coupled between the low-noise amplifier and the hybrid filter.

18. The radio-frequency module according to claim 13, wherein the power amplifier is disposed at the first major surface.

19. A communication device comprising: a radio-frequency (RF) signal processing circuit configured to process a radio-frequency signal received by an antenna; and the radio-frequency module according to claim 10, the radio-frequency module being configured to transfer the radio-frequency signal between the antenna and the RF signal processing circuit.
